```
while not eof(file) do
  x ← read(file)
  total ← x
```

The code above abstract reading integers from a file and summing them, until we reach the end of the file (`eof`).

For a more concrete example, consider the algorithm to compute the greatest common divisor (*GCD*) of two integers a and b:

```
while a <> b do
  if a > b then
    a ← a – b
  else
    b ← b - a
```

At the end of the loop, variable `a` will hold the GCD of `a` and `b`.

As you can see, both examples need to evaluate a predicate that is clearly harder to express with a for loop.

A.4.3 Break and Continue

Sometimes you need to check multiple conditions or react to conditions that can only be evaluated inside the body of the loops. To cope with that, we can use `break` and `continue`.